2.44 Proof by Contradiction

Worked Solutions 1–18

1)

a) The cat is not black
b) No cats are black
c) All cats are black
d) Line A is not parallel to line B
e) The probability has not increased
f) $x < 0$
g) $x \neq 2$
h) a is rational
i) n is odd and not prime

2)

a) There exists an integer n that is neither positive nor negative (e.g. $n = 0$)
b) n is a number with less than 3 distinct prime factors
c) There exists at least one prime number that is not odd (e.g. 2)
d) There exists a rational number and an irrational number whose sum is rational.
e) If n^2 is even, then n is odd

3)

Given that p and q are positive integers, assume that $\frac{p}{q} + \frac{q}{p} < 2$

4)

a) Assume that $\sqrt{2}$ is rational, so $\sqrt{2} = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$$

Hence a^2 is even, so a is even. Let $a = 2k$.

$$(2k)^2 = 4k^2 = 2b^2 \Rightarrow 2k^2 = b^2$$

Hence b^2 is even, so b is even. This contradicts that $\frac{a}{b}$ is in its simplified form. Therefore $\sqrt{2}$ is irrational.

b) Assume that $\sqrt{3}$ is rational, so $\sqrt{3} = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$3 = \frac{a^2}{b^2} \Rightarrow a^2 = 3b^2$$

Hence a^2 is divisible by 3, so a is divisible by 3. Let $a = 3k$.

$$(3k)^2 = 9k^2 = 3b^2 \Rightarrow 3k^2 = b^2$$

Hence b^2 is divisible by 3, so b is divisible by 3. This contradicts that $\frac{a}{b}$ is in its simplified form. Therefore $\sqrt{3}$ is irrational.

c) Assume that $\sqrt{5}$ is rational, so $\sqrt{5} = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$5 = \frac{a^2}{b^2} \Rightarrow a^2 = 5b^2$$

Hence a^2 is divisible by 5, so a is divisible by 5. Let $a = 5k$.

$$(5k)^2 = 25k^2 = 5b^2 \Rightarrow 5k^2 = b^2$$

Hence b^2 is divisible by 5, so b is divisible by 5. This contradicts that $\frac{a}{b}$ is in its simplified form. Therefore $\sqrt{5}$ is irrational.

5)

Assume that $\sqrt[3]{5}$ is rational, so $\sqrt[3]{5} = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$5 = \frac{a^3}{b^3} \Rightarrow a^3 = 5b^3$$

Hence a^3 is divisible by 5, so a is divisible by 5. Let $a = 5k$.

$$(5k)^3 = 125k^3 = 5b^3 \Rightarrow 125k^3 = b^3$$

Hence b^3 is divisible by 5, so b is divisible by 5. This contradicts that $\frac{a}{b}$ is in its simplified form. Therefore $\sqrt[3]{5}$ is irrational.

6)

Assume that $1 + \sqrt{2}$ is rational, so $1 + \sqrt{2} = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$\sqrt{2} = \frac{a}{b} - 1 \Rightarrow \sqrt{2} = \frac{a-b}{b}$$

$$2 = \left(\frac{a-b}{b}\right)^2 \Rightarrow (a-b)^2 = 2b^2$$

Hence $(a-b)^2$ is even, so $a - b$ is even. Let $a - b = 2k$.

$$(2k)^2 = 4k^2 = 2b^2 \Rightarrow 2k^2 = b^2$$

Hence b^2 is even, so b is even. This contradicts that $\frac{a}{b}$ is in its simplified form. Therefore $1 + \sqrt{2}$ is irrational.

7)

Assume that if $x^2 - 5x + 6 < 0$ then $x \leq 2$ or $x \geq 3$.

$$x^2 - 5x + 6 = (x-2)(x-3)$$

So we are assuming that $(x-2)(x-3) < 0$. This has solutions $2 < x < 3$ which contradicts $x \leq 2$ or $x \geq 3$. Therefore if $x^2 - 5x + 6 < 0$ then $2 < x < 3$.

8)

Assume that there is a greatest multiple of 3, $3k$. Consider $3(k+1) = 3k + 3$ which is both a multiple of 3 and greater than $3k$ which contradicts the assumption. Therefore there is no greatest multiple of 3.

9)

Assume that there is a largest prime number, p_n. Let $N = p_1 p_2 \dots p_n$ where $p_1, p_2, \dots, p_n$ are all the prime numbers. Consider $N + 1 = p_1 p_2 \dots p_n + 1$ which is either prime or has a prime factor not in the list, which is a contradiction. Therefore there are infinitely many prime numbers.

10)

Assume that the difference between a rational number and an irrational number is rational. Let a, b, p and q be integers. Let $\frac{a}{b}$ be rational ($b \neq 0$) and c be irrational. So assume that $\frac{a}{b} - c = \frac{p}{q}$.

$$c = \frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{bp}{bq} = \frac{aq - bp}{bq}$$

Which contradicts c being irrational. Therefore the difference between a rational number and an irrational number is always irrational.

11)

a) Assume that the product of a rational number and an irrational number is rational. Let a, b, p and q be integers. Let $\frac{a}{b}$ be rational ($b \neq 0$) and c be irrational. So assume that $\frac{a}{b} \times c = \frac{p}{q}$.

$$c = \frac{p}{q} \div \frac{a}{b} = \frac{p}{q} \times \frac{b}{a} = \frac{bp}{aq}$$

Which contradicts c being irrational. Therefore the product of a positive rational and a positive irrational number is irrational.

b) It is true for all cases except 0.

12)

There are no real solutions when $b^2 - 4ac < 0$.

$$(-3k)^2 - 4k \times 8 < 0 \Rightarrow 9k^2 - 32k < 0$$

Assume that if $9k^2 - 32k < 0$ then $k \leq 0$ or $k \geq \frac{32}{9}$.

$$k \leq 0 \text{ or } k \geq \frac{32}{9} \Rightarrow k(9k - 32) \geq 0 \Rightarrow 9k^2 - 32k \geq 0$$

Which contradicts that $9k^2 - 32k < 0$. Therefore if $kx^2 - 3kx + 8$ has no real roots, then $9k^2 - 32k < 0$ and so $0 < k < \frac{32}{9}$.

13)

Assume that for all $x > 0$ then $x + \frac{1}{x} < 2$.

$$\frac{x^2}{x} + \frac{1}{x} < 2 \Rightarrow \frac{x^2 + 1}{x} < 2 \Rightarrow x^2 + 1 < 2x$$

$$x^2 - 2x + 1 < 0 \Rightarrow (x - 1)^2 < 0$$

But anything squared is ≥ 0 so this is a contradiction. Therefore for all $x > 0$ then $x + \frac{1}{x} \geq 2$.

14)

Assume that for any value of a and b then $a^2 + b^2 < 2ab$.

$$a^2 - 2ab + b^2 < 0 \Rightarrow (a - b)(a - b) < 0 \Rightarrow (a - b)^2 < 0$$

But anything squared is ≥ 0 so this is a contradiction. Therefore $a^2 + b^2 \geq 2ab$.

15)

Assume that $\log_{10} 3$ is a rational number. So $\log_{10} 3 = \frac{a}{b}$ where a and b are positive integers, b is non-zero, and $\frac{a}{b}$ is in its simplified form.

$$\log_{10} 3 = \frac{a}{b} \Rightarrow 3 = 10^{a/b} \Rightarrow 3^b = (10^{a/b})^b = 10^a$$

But powers of 3 are odd and powers of 10 are even, therefore this contradicts $\log_{10} 3 = \frac{a}{b}$. Therefore $\log_{10} 3$ is irrational.

16)

a) $p^2 - 6q = 5 \Rightarrow p^2 = 6q + 5$. $6q$ is even, 5 is odd, even + odd is odd. Therefore p^2 is odd.

b) Assume integers a and b exist such that $a^2 - 6b = 5$. From part a) we have that a is odd. Therefore $a = 2k + 1$ where k is an integer.

$$(2k+1)^2 - 6b = 5 \Rightarrow 4k^2 + 4k + 1 = 6b + 5 \Rightarrow 4k^2 + 4k - 4 = 6b$$

$$2k^2 + 2k - 2 - 3b = 0$$

There must exist a solution to $2k^2 + 2k - 2 - 3b = 0$ such that b is an integer:

$$2k^2 + 2k - (2 + 3b) = 0$$

$$k = \frac{2 \pm \sqrt{4 + 8(2 + 3b)}}{4} = -\frac{1}{2} \pm \frac{\sqrt{5 - 6b}}{2}$$

Since k is an integer, $\sqrt{5 - 6b}$ must exist and be an integer. Therefore $5 - 6b > 0$ and $5 - 6b$ must be an integer.

$$5 - 6b > 0 \Rightarrow b < \frac{5}{6}$$

Given b is an integer, there are no possible values of b such that $5 - 6b$ is a positive integer, which contradicts the statement that integers a and b exist such that $a^2 - 6b = 5$. Therefore there are no integers a and b such that $a^2 - 6b = 5$.

17)

Assume that there exists a prime number p which is 1 less than a square number and $p \neq 3$.

$$p = n^2 - 1 \text{ where } p \neq 3$$

$$p = (n + 1)(n - 1)$$

Since p is prime and must have precisely two factors, $p = 1 \times (n - 1)$ or $p = (n + 1) \times 1$. Either $n + 1 = 1$ or $n - 1 = 1$.

Case 1: $n + 1 = 1 \Rightarrow n = 0$. Therefore $p = -1$ which is not a prime.

Case 2: $n - 1 = 1 \Rightarrow n = 2$. Therefore $p = 4 - 1 = 3$ which contradicts that $p \neq 3$. Therefore 3 is the only prime number which is 1 less than a square number.

18)

Assume that $\cos\theta + \sin\theta > \sqrt{2}$.

$$\begin{aligned}
(\cos\theta + \sin\theta)^2 &> 2 \\
\cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta &> 2 \\
2\sin\theta\cos\theta + 1 &> 2 \\
2\sin\theta\cos\theta &> 1 \\
\sin 2\theta &> 1
\end{aligned}$$

But $1 \leq \sin 2\theta \leq 1$ which is a contradiction. Therefore $\cos\theta + \sin\theta \leq \sqrt{2}$.